

Filters, Transfer Functions & Anti-Aliasing

A Comprehensive Guide for GNC Engineers

March 2026

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1 What is a Filter?

A filter is a system that **selectively allows certain frequencies** of a signal to pass through while **attenuating (reducing) others**. Think of it like a sieve—but for frequencies instead of particle sizes.

2 Low-Pass Filter (LPF)

A low-pass filter **passes low frequencies and blocks high frequencies**.

- **Below the cutoff frequency f_c :** signal passes through (mostly unchanged).
- **Above the cutoff frequency f_c :** signal is attenuated (weakened).

Use cases: Smoothing noisy sensor data, removing high-frequency vibration from a control signal, anti-aliasing before sampling.

Analogy: A heavy blanket on a drum—slow (low-frequency) movements pass, but fast vibrations are absorbed.

3 High-Pass Filter (HPF)

A high-pass filter **passes high frequencies and blocks low frequencies**. The opposite of a low-pass filter.

- **Above f_c :** signal passes.
- **Below f_c :** signal is attenuated.

Use cases: Removing DC offset or slow drift from a signal, extracting fast-changing dynamics.

Analogy: A stiff spring—it responds to fast jerky inputs but ignores slow steady pushes.

4 Fundamentals of the Cutoff Frequency

4.1 Definition

The cutoff frequency f_c (also called **corner frequency**, **break frequency**, or **−3 dB frequency**) is the frequency at which a filter's output power drops to **half** of its pass-band (maximum) power. Equivalently, the output amplitude drops to $\frac{1}{\sqrt{2}} \approx 0.707$ of the passband amplitude:

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}} \iff -3 \text{ dB} \quad (1)$$

4.2 What It Physically Represents

The cutoff frequency is the **boundary between two regions**:

Region	Frequency Range	What Happens
Passband	$f < f_c$ (for low-pass)	Signal passes with little or no attenuation
Stopband	$f > f_c$ (for low-pass)	Signal is progressively attenuated

f_c is not a hard wall—it is a **transition point**. The filter does not suddenly switch from “pass” to “block” at f_c . Instead, attenuation increases gradually, and f_c marks where it becomes significant.

4.3 Two Equivalent Ways to Express It

1. **In frequency (Hz):** f_c [Hz]
2. **In angular frequency (rad/s):** $\omega_c = 2\pi f_c$ [rad/s]

Transfer functions typically use ω_c because the math is cleaner. Datasheets and practical specifications typically use f_c .

4.4 How It Relates to the Transfer Function

For a first-order low-pass filter:

$$H(s) = \frac{\omega_c}{s + \omega_c} = \frac{1}{\tau s + 1} \quad (2)$$

The cutoff frequency is directly encoded in the denominator. The pole is at $s = -\omega_c$, and:

$$\tau = \frac{1}{\omega_c} = \frac{1}{2\pi f_c} \quad (3)$$

where τ is the **time constant**—the time-domain equivalent of f_c .

4.5 Relationship Between f_c and Time Constant τ

These are two sides of the same coin:

$$f_c = \frac{1}{2\pi\tau} \quad \Longleftrightarrow \quad \tau = \frac{1}{2\pi f_c} \quad (4)$$

Parameter	Domain	Meaning
f_c	Frequency domain	Frequency where attenuation starts
τ	Time domain	Time for step response to reach 63%

- **Higher** f_c = smaller τ = faster response, lets more frequencies through.
- **Lower** f_c = larger τ = slower response, heavier smoothing.

4.6 Behaviour at Three Key Frequencies

For a single-pole low-pass filter:

Frequency	Gain $ H $	Gain (dB)	Phase Shift	Physical Meaning
$f \ll f_c$	≈ 1	≈ 0 dB	≈ 0	Signal passes unchanged
$f = f_c$	$\frac{1}{\sqrt{2}}$	-3 dB	-45	Half-power point (cutoff)
$f \gg f_c$	$\approx \frac{f_c}{f} \rightarrow 0$	$\rightarrow -\infty$	$\rightarrow -90$	Signal strongly attenuated

4.7 How to Choose the Cutoff Frequency

The choice of f_c depends on the application.

Rule of thumb: Set f_c above the highest frequency you care about, but below the frequencies you want to reject:

$$f_{\text{signal, max}} < f_c < f_{\text{noise, min}} \quad (5)$$

Example: Useful signal occupies 0–20 Hz, noise starts around 60 Hz. Set $f_c \approx 30$ –40 Hz. This preserves the signal while attenuating the noise.

If f_c is too high...	If f_c is too low...
Noise passes through	Useful signal gets attenuated
Poor filtering	Slow response, phase lag
Signal is clean but noisy	Signal is smooth but distorted/delayed

4.8 Cutoff Frequency in Hardware (RC Circuit)

For a simple RC low-pass filter (resistor + capacitor):

$$f_c = \frac{1}{2\pi RC} \quad (6)$$

- Want **lower** f_c ? Increase R or $C \rightarrow$ more smoothing, slower response.
- Want **higher** f_c ? Decrease R or $C \rightarrow$ less smoothing, faster response.

4.9 Cutoff Frequency in Digital Filters

In a digital (discrete-time) system running at sample rate f_s , the cutoff frequency must satisfy:

$$f_c < \frac{f_s}{2} \quad (7)$$

A common discrete approximation of a first-order low-pass filter is the **exponential moving average**:

$$y[n] = \alpha \cdot x[n] + (1 - \alpha) \cdot y[n - 1] \quad (8)$$

where the smoothing factor α relates to f_c by:

$$\alpha = \frac{2\pi f_c \Delta t}{1 + 2\pi f_c \Delta t} = \frac{2\pi f_c / f_s}{1 + 2\pi f_c / f_s} \quad (9)$$

- $\alpha \rightarrow 1$: no filtering (output = input).
- $\alpha \rightarrow 0$: heavy filtering (output barely changes).

4.10 Cutoff Frequency on a Bode Plot

On a Bode magnitude plot, f_c is located where the two straight-line asymptotes—the flat passband and the -20 dB/decade slope—**intersect**. This is why f_c is also called the “corner frequency.”

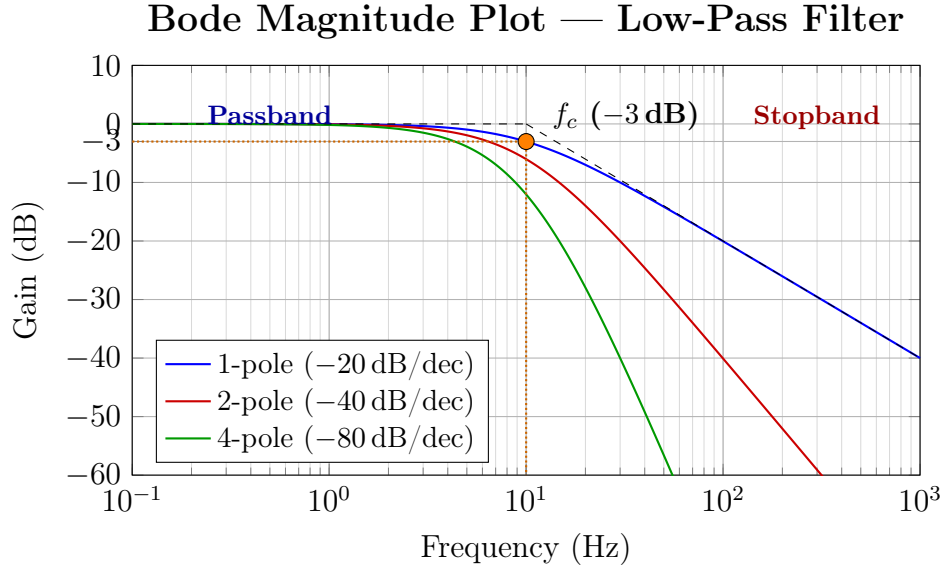


Figure 1: Bode magnitude plot of low-pass filters with $f_c = 10$ Hz. The orange dot marks the -3 dB cutoff point. Higher-order (multi-pole) filters have steeper roll-off in the stopband.

Cutoff Frequency Summary

Concept	Key Fact
What	Frequency where output power = half of input power
Value	$ H = 1/\sqrt{2} \approx 0.707$ (-3 dB)
Time-domain eq.	$\tau = 1/(2\pi f_c)$
In the TF	Equals the pole location: ω_c
On a Bode plot	The corner where the two asymptotes meet
Design rule	Set above signal bandwidth, below noise frequencies

5 Why -3 dB?

The -3 dB point comes from a specific physical meaning: it is where the output **power** is exactly **half** of the input power.

5.1 The Power Definition

Decibels for power are defined as:

$$\text{dB} = 10 \log_{10} \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right) \quad (10)$$

At half power:

$$10 \log_{10}\left(\frac{1}{2}\right) = 10 \times (-0.3010) = -3.010 \text{ dB} \quad (11)$$

5.2 Amplitude Interpretation

Since power is proportional to amplitude squared ($P \propto V^2$), half power corresponds to an amplitude ratio of:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}} \approx 0.707 \quad (12)$$

Verification using the amplitude dB formula:

$$20 \log_{10}\left(\frac{1}{\sqrt{2}}\right) = 20 \times (-0.1505) = -3.01 \text{ dB} \quad (13)$$

5.3 Why Half-Power is the Natural Breakpoint

For a single-pole low-pass filter:

$$H(s) = \frac{\omega_c}{s + \omega_c} \quad (14)$$

Evaluate the magnitude at $\omega = \omega_c$:

$$|H(j\omega_c)| = \frac{\omega_c}{\sqrt{\omega_c^2 + \omega_c^2}} = \frac{\omega_c}{\omega_c \sqrt{2}} = \frac{1}{\sqrt{2}} \quad (15)$$

At the frequency where ω **equals** the pole location ω_c , the gain is always $\frac{1}{\sqrt{2}}$. This is the natural “corner” of the filter—where the two asymptotes of the Bode plot intersect. Half-power (−3 dB) is not an arbitrary convention; it is the **inherent mathematical breakpoint** of the filter’s transfer function.

Summary Table

Fact	Value
−3 dB in power	$P_{\text{out}}/P_{\text{in}} = 1/2$ (half power)
−3 dB in amplitude	$V_{\text{out}}/V_{\text{in}} = 1/\sqrt{2} \approx 0.707$
Why this value?	$\omega = \omega_c$ in the transfer function—the natural corner

6 Transfer Function and Its Relation to Filters

6.1 What is a Transfer Function?

A transfer function $H(s)$ is a **mathematical representation** of a system’s input–output relationship in the **Laplace domain** (s-domain):

$$H(s) = \frac{Y(s)}{X(s)} \quad (16)$$

where:

- $Y(s)$ = output (Laplace transform),
- $X(s)$ = input (Laplace transform),
- $s = \sigma + j\omega$ is the complex frequency variable.

6.2 How It Relates to Filters

Every filter IS a transfer function. The transfer function defines *how* each frequency is treated:

Filter Type	Transfer Function	Behavior
Low-Pass	$H(s) = \frac{\omega_c}{s + \omega_c}$	Gain $\rightarrow 1$ at low freq, $\rightarrow 0$ at high freq
High-Pass	$H(s) = \frac{s}{s + \omega_c}$	Gain $\rightarrow 0$ at low freq, $\rightarrow 1$ at high freq

where $\omega_c = 2\pi f_c$ is the cutoff frequency in rad/s.

6.3 Frequency Response

To see how a filter behaves at a specific frequency, substitute $s = j\omega$.

Low-Pass example:

$$H(j\omega) = \frac{\omega_c}{j\omega + \omega_c} \quad (17)$$

The magnitude is:

$$|H(j\omega)| = \frac{\omega_c}{\sqrt{\omega^2 + \omega_c^2}} \quad (18)$$

- When $\omega \ll \omega_c$: $|H| \approx 1$ (signal passes)
- When $\omega = \omega_c$: $|H| = \frac{1}{\sqrt{2}}$ (-3 dB point)
- When $\omega \gg \omega_c$: $|H| \approx \frac{\omega_c}{\omega} \rightarrow 0$ (signal blocked)

7 Single-Pole vs. Multi-Pole Low-Pass Filters

7.1 What is a “Pole”?

A **pole** is a root of the **denominator** of the transfer function. It determines how aggressively the filter rolls off (attenuates) at high frequencies.

In $H(s) = \frac{\omega_c}{s + \omega_c}$, the denominator $s + \omega_c = 0$ gives the pole at $s = -\omega_c$.

7.2 Single-Pole (First-Order) Low-Pass Filter

Transfer function:

$$H(s) = \frac{\omega_c}{s + \omega_c} = \frac{1}{\frac{s}{\omega_c} + 1} = \frac{1}{\tau s + 1} \quad (19)$$

where $\tau = \frac{1}{\omega_c}$ is the **time constant**.

Key Characteristics

Property	Value
Roll-off rate	−20 dB/decade (−6 dB/octave)
Phase shift at f_c	−45
Maximum phase shift	−90
Number of poles	1

Roll-off meaning: Above the cutoff, every time the frequency increases by $10\times$, the output drops by 20 dB (a factor of 10 in amplitude).

Time domain: The step response is an exponential rise:

$$y(t) = 1 - e^{-t/\tau} \quad (20)$$

It reaches $\sim 63\%$ of the final value after one time constant τ .

Pros: Simple, no overshoot, always stable.

Cons: Gentle roll-off—doesn't reject high frequencies very sharply.

7.3 Multi-Pole (Higher-Order) Low-Pass Filters

Adding more poles makes the filter **steeper** (sharper cutoff).

7.3.1 Second-Order (2-Pole) Low-Pass Filter

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (21)$$

where:

- ω_n = natural frequency,
- ζ = damping ratio (controls overshoot/ringing).

Roll-off: −40 dB/decade (twice as steep).

7.3.2 N -th Order (N -Pole) Low-Pass Filter

General form:

$$H(s) = \frac{\omega_c^N}{\prod_{i=1}^N (s + p_i)} \quad \text{or} \quad H(s) = \frac{1}{(\tau s + 1)^N} \text{ (Butterworth-like)} \quad (22)$$

Order (Poles)	Roll-off Rate	Phase at f_c
1	−20 dB/decade	−45
2	−40 dB/decade	−90
3	−60 dB/decade	−135
N	−20 N dB/decade	− $N \times 45$

7.4 Trade-offs of More Poles

	Single-Pole	Multi-Pole
Roll-off	Gentle (−20 dB/dec)	Sharp (−20 N dB/dec)
Phase lag	Less (max −90)	More (max − $N \times 90$)
Stability	Always stable	Can oscillate if poorly tuned
Overshoot	None	Possible (depends on ζ)
Delay	Less	More group delay
Complexity	Simple	More computation/components

8 What Does “A Signal Contains Frequencies” Mean?

8.1 Fourier’s Theorem

Any real-world signal—no matter how complex—can be decomposed into a **sum of sine waves** at different frequencies. This is Fourier’s theorem.

For example, a vibration sensor output might be:

$$x(t) = \underbrace{3 \sin(2\pi \cdot 10 t)}_{10 \text{ Hz component}} + \underbrace{1.5 \sin(2\pi \cdot 50 t)}_{50 \text{ Hz component}} + \underbrace{0.2 \sin(2\pi \cdot 300 t)}_{300 \text{ Hz component}} + \dots \quad (23)$$

This signal “contains” frequencies of 10 Hz, 50 Hz, 300 Hz, etc. Each is a building block of the total signal.

9 Digitizing a Signal

9.1 The Real World is Continuous

A physical signal—like the voltage from a sensor—exists **continuously** in time. At every instant it has a value. This is an **analog signal**: smooth, unbroken, and existing in nature.

9.2 Computers Need Discrete Numbers

A computer cannot store infinitely many values. To get a real-world signal into a computer, you must convert it from analog to digital. This is **digitizing**.

9.3 How Digitizing Works

A device called an **ADC (Analog-to-Digital Converter)** performs two steps:

1. **Sampling** — Measure the signal's value at regular intervals, spaced $T_s = 1/f_s$ seconds apart. If $f_s = 1000$ Hz, you take 1000 measurements per second, one every 1 ms.
2. **Quantization** — Round each measurement to a finite precision. The continuous voltage (e.g., 2.3471928... V) gets rounded to the nearest representable value (e.g., 2.347 V with a 12-bit ADC).

Key Terms

Term	Meaning
Analog signal	Continuous—exists at every instant in time
Digitize	Convert into a sequence of numbers a computer can store
Sampling	Measuring the signal at fixed time intervals
Sample rate f_s	How many measurements per second
ADC	The hardware that performs this conversion

10 Anti-Aliasing

10.1 The Problem: Aliasing

When you **sample** a continuous signal at discrete time intervals, you are only capturing snapshots. If the signal contains frequencies **higher than half your sampling rate**, those high frequencies get misidentified as **lower frequencies** in the sampled data.

This is **aliasing**—a high-frequency signal masquerading as a low-frequency one. Once it happens, it **cannot be undone**.

10.2 The Nyquist–Shannon Theorem

To correctly capture a frequency f , you must sample at a rate:

$$f_s > 2f \quad (24)$$

The critical boundary $f_s/2$ is called the **Nyquist frequency** f_N :

$$f_N = \frac{f_s}{2} \quad (25)$$

- Frequencies **below** f_N : correctly captured.
- Frequencies **above** f_N : aliased (folded back into the 0 to f_N range).

10.3 Aliasing Example

Consider sampling a 90 Hz sine wave at $f_s = 100$ Hz:

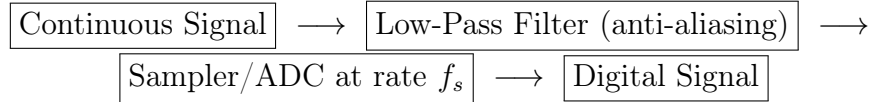
- Nyquist frequency: $f_N = 50$ Hz
- 90 Hz > 50 Hz $\Rightarrow$ **aliased**

- The 90 Hz signal appears as a 10 Hz signal: $f_s - 90 = 10$ Hz

From the sampled data alone, you cannot distinguish whether the original was 90 Hz or 10 Hz. They produce identical sample values.

10.4 The Solution: Anti-Aliasing Filter

An **anti-aliasing filter** is a **low-pass filter** placed **before** the sampler/ADC that removes all frequencies above f_N so they cannot alias:



11 How the Anti-Aliasing Filter Works

The filter does not “detect and remove” aliased frequencies. It works **before sampling** by **attenuating high-frequency components** so that by the time the ADC samples the signal, those dangerous frequencies are too small to matter. It is **prevention**, not cure.

11.1 Concrete Example

Setup: Sample at $f_s = 100$ Hz, so $f_N = 50$ Hz.

Raw analog signal contains:

Component	Frequency	Amplitude
Useful signal	5 Hz	3.0 V
Useful signal	20 Hz	1.5 V
Noise/vibration	90 Hz	0.8 V
Noise/vibration	300 Hz	0.4 V

11.1.1 Without Anti-Aliasing Filter

Original	Appears as	Status
5 Hz	5 Hz	Correct
20 Hz	20 Hz	Correct
90 Hz	10 Hz	ALIASED ($100 - 90 = 10$)
300 Hz	0 Hz	ALIASED (folds multiple times)

11.1.2 With Anti-Aliasing Filter ($f_c \approx 40$ Hz, 2nd-order)

Component	Freq.	Original	Filter Gain	After Filter
Useful	5 Hz	3.0 V	≈ 1.00 (-0.1 dB)	3.0 V ✓
Useful	20 Hz	1.5 V	≈ 0.97 (-0.3 dB)	1.46 V ✓
Noise	90 Hz	0.8 V	≈ 0.19 (-14 dB)	0.15 V ↓
Noise	300 Hz	0.4 V	≈ 0.018 (-35 dB)	0.007 V ↓↓

The dangerous frequencies are now tiny. When the ADC samples, they are too small to cause meaningful corruption.

11.2 Physical Mechanism: RC Filter

The simplest anti-aliasing filter is an RC low-pass filter (a resistor and capacitor):

- At **low frequencies**: the signal changes slowly. The capacitor has time to fully charge and discharge, so the output follows the input. Signal passes through.
- At **high frequencies**: the signal changes so fast that the capacitor **cannot keep up**. It acts as a short circuit to ground, bleeding the high-frequency energy away.

The cutoff frequency is:

$$f_c = \frac{1}{2\pi RC} \quad (26)$$

11.3 Why Steeper Filters Are Better

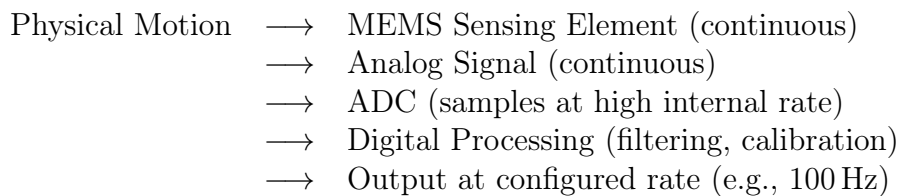
A real filter cannot transition from gain = 1 to gain = 0 instantly at f_c . There is always a **transition band** between f_c and $f_s/2$. A steeper (multi-pole) filter makes this transition band narrower:

Filter Order	Roll-off	Attenuation at $2 \times f_c$
1-pole	−20 dB/dec	−6 dB (amplitude $\times 0.5$)
2-pole	−40 dB/dec	−12 dB (amplitude $\times 0.25$)
4-pole	−80 dB/dec	−24 dB (amplitude $\times 0.06$)

12 MEMS IMU and Internal Signal Processing

A MEMS IMU senses **continuous** physical motion (acceleration, angular rate) at all times, but internally it **digitizes** that signal and outputs discrete data packets at the configured output data rate (e.g., 100 Hz).

12.1 Internal Processing Chain



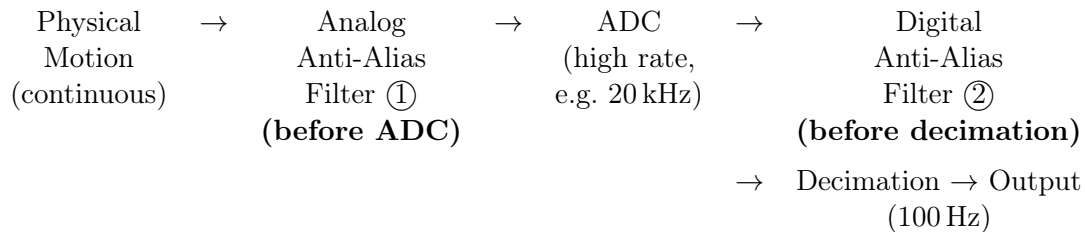
12.2 Typical Internal Rates

Stage	Typical Rate
Internal ADC sampling	4 kHz – 32 kHz
Internal anti-aliasing filter	Applied before decimation
Output data rate (what you receive)	100 Hz (often configurable)

13 Two Stages of Anti-Aliasing in an IMU

There are **two** anti-aliasing steps at two different points in the processing chain.

13.1 The Full Processing Chain



13.2 Stage ①: Analog Anti-Aliasing Filter (Before ADC)

This is a **hardware analog filter** that sits before the ADC to prevent aliasing at the **internal** sampling rate.

- Internal ADC samples at, say, 20 kHz $\Rightarrow$ Nyquist = 10 kHz.
- The analog filter removes frequencies above ~ 10 kHz.
- Because the internal rate is so high, a simple 1st- or 2nd-order analog filter is sufficient—there is a huge margin between the signal bandwidth (say 0–500 Hz) and 10 kHz.

13.3 Stage ②: Digital Anti-Aliasing Filter (After ADC, Before Decimation)

After the ADC, data exists at 20 kHz. But the output is only 100 Hz. Going from 20 kHz to 100 Hz is called **decimation**—keeping only every 200th sample.

Decimation is a form of re-sampling. And just like any sampling, it can cause aliasing if you do not filter first.

- New output rate: 100 Hz $\Rightarrow$ New Nyquist: 50 Hz.
- Must remove everything above 50 Hz **before** decimating.
- This filter operates on **digital data** (computed in firmware, not hardware).

13.4 Summary of Both Stages

Stage	What's being “sampled”	Filter Type	Prevents aliasing from
① Before ADC	Continuous $\rightarrow$ 20 kHz digital	Analog (hardware)	Freq. above 10 kHz folding
② Before decimation	20 kHz digital $\rightarrow$ 100 Hz digital	Digital (firmware)	Freq. above 50 Hz folding

The Fundamental Rule

An anti-aliasing filter must be applied **before** sampling. This rule is never violated:

- Stage ① is **before** the ADC samples the analog signal.
- Stage ② is **before** the decimation re-samples the digital signal.

Every time you reduce the sample rate—whether analog→digital or high-rate digital→low-rate digital—you need a low-pass filter **before** that rate reduction.

14 Practical Summary

- Use a **single-pole LPF** when you need simple smoothing and can tolerate some high-frequency leakage—common for quick sensor filtering in GNC systems.
- Use a **multi-pole LPF** when you need a sharp cutoff between pass and stop bands—e.g., anti-aliasing filters, vibration isolation, or strict bandwidth control.
- The **transfer function** is the mathematical DNA of the filter—it tells you everything about gain, phase, stability, and transient behavior at every frequency.
- **Anti-aliasing** is prevention, not cure. Once aliasing occurs, the corrupted data is irrecoverable.
- In a MEMS IMU, anti-aliasing happens at **two stages**: an analog filter before the internal ADC, and a digital filter before decimation to the output rate.